THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4 : 1	Convert to the same unit
2	**(c)** 4 : 6 : 5	Make the middle terms equal
3	**(c)** 17.5 km	3.5×5
4	**(d)** 10 days	Fewer or more workers: days vary inversely
5	**(a)** 420 km	Speed $240 \div 4$
6	**(b)** ₹ 150	One part $= 150$
7	**(b)** decreases	Inverse proportion
8	**(b)** 5 : 1	Convert to the same unit
9	**(b)** 6 : 8 : 7	Make the middle terms equal
10	**(b)** 48 km	4×12

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 m 60 cm $= 360$ cm (1); ratio $= 4 : 1$ (1).	1+1
12	Total work $= 9 \times 16 = 144$ worker-days (1); days $= 144 \div 12 = 12$ (1).	1+1
13	3.5 cm $\rightarrow$ 70 km (1); 100 km $\rightarrow 5$ cm (1).	1+1
14	2 m 10 cm $= 210$ cm (1); ratio $= 3 : 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 4 : 6 : 5$ (2).	1+2
16	Man-days $= 10 \times 18 = 180$ (1); days for 15 men $= 180 \div 15$ (1); 12 days (1).	1+1+1
17	One part $= 6$ (1); present ages 18 and 30 (1); after 4 years 22 and 34 (1).	1+1+1
	OR: Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 6 : 8 : 7$ (2).	1+2
18	Man-days $= 12 \times 25 = 300$ (1); days for 20 men $= 300 \div 20$ (1); 15 days (1).	1+1+1
	OR: One part $= 12$ (1); present ages 24 and 36 (1); after 5 years 29 and 41 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	One part ₹ 150 (1); shares ₹ 150, ₹ 450, ₹ 600 (2); after the transfer ₹ 250, ₹ 450, ₹ 500 (1); new ratio $5 : 9 : 10$ (1).	1+2+1+1
	OR: (i) speed $= 240 \div 4 = 60$ km/h (1.5); (ii) $60 \times 7 = 420$ km (1.5); (iii) $360 \div 60 = 6$ hours (2).	1½+1½+2
20	One part ₹ 200 (1); shares ₹ 400, ₹ 600, ₹ 800 (2); after the transfer ₹ 600, ₹ 600, ₹ 600 (1); new ratio $1 : 1 : 1$ (1).	1+2+1+1
	OR: (i) speed $= 180 \div 3 = 60$ km/h (1.5); (ii) $60 \times 5 = 300$ km (1.5); (iii) $270 \div 60 = 4.5$ hours (2).	1½+1½+2

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 15 days	Fewer or more workers: days vary inversely
2	**(a)** 300 km	Speed $180 \div 3$
3	**(b)** ₹ 400	One part $= 200$
4	**(d)** direct proportion	Double the articles, double the cost
5	**(a)** 4 : 1	Convert to the same unit
6	**(a)** 3 : 6 : 5	Make the middle terms equal
7	**(b)** 52 km	6.5×8
8	**(b)** 16 days	Fewer or more workers: days vary inversely
9	**(c)** 528 km	Speed $330 \div 5$
10	**(b)** ₹ 600	One part $= 200$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total work $= 14 \times 15 = 210$ worker-days (1); days $= 210 \div 10 = 21$ (1).	1+1
12	4.2 cm $\rightarrow$ 126 km (1); 150 km $\rightarrow$ 5 cm (1).	1+1
13	5 m 40 cm $= 540$ cm (1); ratio $= 3 : 1$ (1).	1+1
14	Total work $= 20 \times 9 = 180$ worker-days (1); days $= 180 \div 15 = 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make B equal: multiply first ratio by 3, second by 1 (1); A : B : C $= 3 : 6 : 5$ (2).	1+2
16	Man-days $= 8 \times 30 = 240$ (1); days for 12 men $= 240 \div 12$ (1); 20 days (1).	1+1+1
17	One part $= 3$ (1); present ages 12 and 21 (1); after 6 years 18 and 27 (1).	1+1+1
	OR: Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 8 : 10 : 9$ (2).	1+2
18	Man-days $= 14 \times 21 = 294$ (1); days for 7 men $= 294 \div 7$ (1); 42 days (1).	1+1+1
	OR: One part $= 4$ (1); present ages 20 and 24 (1); after 8 years 28 and 32 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	One part ₹ 200 (1); shares ₹ 600, ₹ 800, ₹ 1000 (2); after the transfer ₹ 900, ₹ 800, ₹ 700 (1); new ratio $9 : 8 : 7$ (1).	1+2+1+1
	OR: (i) speed $= 330 \div 5 = 66$ km/h (1.5); (ii) $66 \times 8 = 528$ km (1.5); (iii) $462 \div 66 = 7$ hours (2).	1½+1½+2
20	One part ₹ 100 (1); shares ₹ 200, ₹ 500, ₹ 800 (2); after the transfer ₹ 300, ₹ 500, ₹ 700 (1); new ratio $3 : 5 : 7$ (1).	1+2+1+1
	OR: (i) speed $= 150 \div 2 = 75$ km/h (1.5); (ii) $75 \times 9 = 675$ km (1.5); (iii) $600 \div 75 = 8$ hours (2).	1½+1½+2

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** inverse proportion	Speed x time = distance
2	**(b)** 4 : 1	Convert to the same unit
3	**(c)** 8 : 10 : 9	Make the middle terms equal
4	**(a)** 60 km	2.4×25
5	**(a)** 12 days	Fewer or more workers: days vary inversely
6	**(b)** 675 km	Speed $150 \div 2$
7	**(a)** ₹ 200	One part $= 100$
8	**(d)** the product of the means	$ad = bc$
9	**(c)** 9 : 2	Convert to the same unit
10	**(b)** 6 : 9 : 4	Make the middle terms equal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2.6 cm $\rightarrow$ 130 km (1); 250 km $\rightarrow$ 5 cm (1).	1+1
12	1 m 80 cm $= 180$ cm (1); ratio $= 4 : 1$ (1).	1+1
13	Total work $= 7 \times 24 = 168$ worker-days (1); days $= 168 \div 8 = 21$ (1).	1+1
14	6.5 cm $\rightarrow$ 260 km (1); 200 km $\rightarrow$ 5 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make B equal: multiply first ratio by 3, second by 1 (1); A : B : C $= 6 : 9 : 4$ (2).	1+2
16	Man-days $= 16 \times 15 = 240$ (1); days for 10 men $= 240 \div 10$ (1); 24 days (1).	1+1+1
17	One part $= 9$ (1); present ages 27 and 36 (1); after 3 years 30 and 39 (1).	1+1+1
	OR: Make B equal: multiply first ratio by 2, second by 1 (1); A : B : C $= 10 : 12 : 7$ (2).	1+2
18	Man-days $= 9 \times 20 = 180$ (1); days for 6 men $= 180 \div 6$ (1); 30 days (1).	1+1+1
	OR: One part $= 2$ (1); present ages 14 and 16 (1); after 10 years 24 and 26 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	One part ₹ 200 (1); shares ₹ 800, ₹ 1000, ₹ 1200 (2); after the transfer ₹ 1200, ₹ 1000, ₹ 800 (1); new ratio $6 : 5 : 4$ (1).	1+2+1+1
	OR: (i) speed $= 420 \div 6 = 70$ km/h (1.5); (ii) $70 \times 4 = 280$ km (1.5); (iii) $210 \div 70 = 3$ hours (2).	1½+1½+2
20	One part ₹ 300 (1); shares ₹ 300, ₹ 600, ₹ 1200 (2); after the transfer ₹ 450, ₹ 600, ₹ 1050 (1); new ratio $3 : 4 : 7$ (1).	1+2+1+1
	OR: (i) speed $= 270 \div 3 = 90$ km/h (1.5); (ii) $90 \times 2 = 180$ km (1.5); (iii) $405 \div 90 = 4.5$ hours (2).	1½+1½+2

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